

SOFIA, Bulgaria

WMO: 156140

Lat: 42.695N

Lon: 23.406E

Elev: 531

StdP: 95.11

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.2	-9.8	-15.1	1.1	-10.6	-12.6	1.3	-8.0	10.9	-0.1	9.9	-0.3	1.6	270	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.1	33.1	19.2	31.1	18.9	29.6	18.6	21.1	28.8	20.2	27.5	19.6	26.7	3.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	14.6	23.7	17.9	13.7	23.1	17.1	13.0	22.2	63.4	28.8	60.4	27.8	57.9	26.6	26.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.3	8.1	7.0	-17.2	36.3	3.5	2.3	-19.8	37.9	-21.8	39.3	-23.8	40.6	-26.4	42.3		
			-17.5	22.9	3.5	1.3	-20.0	23.8	-22.1	24.6	-24.0	25.3	-26.5	26.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-0.8	1.9	6.2	11.1	15.8	19.4	21.7	21.8	17.2	11.7	6.2	0.5
	DBStd	8.82	4.79	5.05	4.33	3.87	3.26	3.25	2.83	2.86	3.52	3.97	4.39	4.71
	HDD10.0	1181	336	228	131	31	2	0	0	0	1	29	128	295
	HDD18.3	2958	594	460	377	217	91	25	4	6	61	206	364	552
	CDD10.0	1587	0	2	13	65	181	283	364	365	217	82	14	2
	CDD18.3	323	0	0	0	1	12	58	110	112	28	2	0	0
	CDH23.3	3378	0	0	2	26	148	576	1122	1155	318	31	1	0
	CDH26.7	1212	0	0	0	3	22	181	443	464	96	3	0	0
Wind	WSAvg	2.5	2.4	2.8	3.0	2.8	2.6	2.5	2.5	2.3	2.4	2.3	2.4	2.4
Precipitation	PrecAvg	627	32	35	41	52	74	76	61	52	49	43	43	48
	PrecMax	1071	93	85	116	151	148	162	187	191	198	124	98	169
	PrecMin	334	2	1	2	2	13	8	7	1	2	1	2	1
	PrecStd	150	23	19	22	27	36	40	41	41	41	32	24	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	18.1	22.1	26.2	29.1	33.2	36.1	35.1	32.2	26.2	20.9	16.2
		MCWB	8.6	10.4	11.3	14.5	16.3	19.9	18.8	19.2	17.9	16.0	13.4	10.6
	2%	DB	10.0	14.2	19.2	23.1	26.5	30.8	33.1	33.1	29.2	23.9	18.1	11.9
		MCWB	6.7	8.5	10.4	13.3	16.3	19.4	19.2	19.3	16.9	15.2	12.3	8.2
	5%	DB	7.9	11.8	16.6	20.5	24.9	28.9	31.0	31.1	27.0	21.1	15.2	9.1
		MCWB	5.7	7.2	9.2	12.2	15.9	19.2	19.1	18.7	16.5	14.1	10.9	7.0
	10%	DB	5.6	9.2	14.1	18.8	23.1	26.8	29.1	29.2	24.8	18.9	13.0	7.1
		MCWB	4.1	6.2	8.1	11.5	15.3	18.4	18.9	18.4	16.0	13.4	10.0	5.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	11.1	12.4	15.9	18.9	22.2	21.9	22.0	19.3	17.5	14.2	11.6
		MCDB	12.9	16.9	19.4	23.5	25.6	29.3	30.0	29.8	27.1	23.8	18.9	15.1
	2%	WB	7.2	9.1	10.9	14.3	17.6	20.7	20.9	20.9	18.3	16.0	12.7	8.8
		MCDB	9.5	13.5	17.6	21.1	24.0	27.6	28.7	28.8	25.7	21.7	17.0	11.4
	5%	WB	5.6	7.7	9.7	13.1	16.7	19.8	20.2	20.0	17.6	14.8	11.5	7.2
		MCDB	7.4	11.1	15.5	19.0	22.8	26.4	27.5	27.7	24.4	19.6	15.0	9.2
	10%	WB	4.1	6.3	8.6	12.0	15.8	18.9	19.5	19.2	16.7	13.9	10.2	5.6
		MCDB	5.4	9.0	13.6	16.8	21.6	25.3	26.6	26.9	23.0	18.1	12.8	6.8
Mean Daily Temperature Range	5% DB	MDBR	8.1	9.7	11.3	11.8	11.9	12.4	13.1	13.8	12.7	11.9	9.6	7.6
		MCDBR	11.3	14.5	16.8	15.9	15.1	15.6	16.3	16.9	17.0	16.2	13.8	10.9
		MCWBR	8.2	9.2	9.0	7.8	6.8	6.5	5.9	5.9	6.7	8.5	8.6	8.0
	5% WB	MCDBR	10.6	12.9	14.6	13.6	13.5	13.9	13.9	14.6	14.2	14.0	11.9	9.7
		MCWBR	7.9	8.5	8.2	7.1	6.6	6.4	5.9	5.8	6.3	7.8	7.6	7.3
Clear-Sky Solar Irradiance	taub	0.315	0.343	0.389	0.448	0.438	0.433	0.435	0.446	0.419	0.377	0.340	0.316	
	taud	2.361	2.296	2.221	2.063	2.162	2.215	2.209	2.179	2.229	2.337	2.410	2.394	
	Ebn at Noon	816	847	846	819	840	843	836	813	806	798	780	779	
	Edh at Noon	82	103	125	158	147	140	140	139	123	98	78	73	
All-Sky Solar Radiation	RadAvg	1.60	2.42	3.44	4.47	5.33	6.03	6.25	5.65	4.16	2.87	1.82	1.34	
	RadStd	0.24	0.31	0.37	0.53	0.45	0.52	0.48	0.50	0.34	0.36	0.19	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.80	N/A	N/A	N/A	+0.50	+0.50	-145	-232	+133	+70
(47)	Regional (40 neighbors)	+0.49	N/A	N/A	N/A	+0.38	N/A	N/A	-139	N/A	N/A

Nomenclature: See separate page